

task -1

Installing and Configuring FileZilla Server for FTP Access

Objective

Install FileZilla Server on a Windows computer, create a user account, and configure a shared folder that the user can access through FTP.

Step 1: Install FileZilla Server

1. Download and install **FileZilla Server** on the Windows machine.
2. Run the installer and follow the installation instructions.
3. Choose to start FileZilla Server automatically with Windows if required.
4. After installation, open the **FileZilla Server administration interface**.

Step 2: Create a User Account

1. Open the FileZilla Server administration interface.
2. Go to **Users** or **User Management**.
3. Click **Add** to create a new user.
4. Enter a username, for example:

ftpuser

5. Set a secure password for the account.
6. Save or apply the user settings.

Step 3: Create a Shared Folder

1. Create a folder on the Windows computer, for example:

C:\FTPShare

2. Return to FileZilla Server.
3. Select the newly created user.
4. Open the **Shared folders** or **Mount points** section.
5. Add the folder:

C:\FTPShare

6. Set it as the user's home or shared directory.
7. Assign the required permissions, such as:

- Read
- Write
- Create
- Delete

For basic FTP testing, **Read** and **Write** permissions are usually sufficient.

Step 4: Configure FTP Access

Make sure the FTP server is listening on the configured port, commonly:

Port: 21

Also check that Windows Firewall allows FileZilla Server or inbound FTP connections on the configured port.

To find the Windows computer's IP address, open Command Prompt and run:

```
ipconfig
```

Note the IPv4 address, for example:

192.168.1.10

Step 5: Test the FTP Connection

On another computer or using an FTP client, connect using:

Host: 192.168.1.10

Username: ftpuser

Password: [your configured password]

Port: 21

You can also test from Command Prompt:

```
ftp 192.168.1.10
```

Enter the username and password when prompted.

Step 6: Verify File Access

After connecting successfully:

1. Open the FTP shared folder.
2. Upload a test file.
3. Download the file again to confirm access.

4. Check whether the file appears in:

C:\FTPShare

Conclusion

FileZilla Server was installed on the Windows machine, a user account named ftpuser was created, and the folder C:\FTPShare was configured for FTP access. The FTP connection can be tested using the server's IP address, username, password, and port 21. Successful file upload and download confirm that the FTP server and shared folder are working correctly.

Task -2

Using FileZilla Client to Upload an Image to an FTP Server

Objective

To connect to the FTP server using **FileZilla Client** and upload an image file from the computer to the configured shared folder.

Step 1: Open FileZilla Client

1. Install and open **FileZilla Client** on the computer.
2. Open the **File → Site Manager** option.
3. Click **New Site**.

Step 2: Enter FTP Server Details

Enter the credentials created on the FileZilla Server:

Protocol: FTP - File Transfer Protocol

Host: 192.168.1.10

Port: 21

Encryption: Use plain FTP

Logon Type: Normal

User: ftpuser

Password: [your configured password]

Replace 192.168.1.10 with the actual IP address of your FTP server.

Click **Connect**.

Step 3: Verify the Connection

After connecting successfully, FileZilla Client should display:

- **Local site** — files and folders on your computer.
- **Remote site** — files and folders available on the FTP server.

The remote directory should show the shared FTP folder configured on the server.

Step 4: Select an Image File

In the **Local site** section:

1. Navigate to the folder containing an image.
2. Select an image, for example:

C:\Users\Student\Pictures\sample.jpg

Step 5: Upload the Image

1. Right-click the selected image.
2. Choose **Upload**.
3. FileZilla will transfer the image to the remote FTP shared folder.
4. Wait until the transfer appears as successful in the **Successful transfers** section.

Step 6: Verify the Uploaded File

Check the **Remote site** section and confirm that the image file is visible there.

For example:

sample.jpg

The file should also appear in the FTP server's shared folder, such as:

C:\FTPShare\sample.jpg

Task-3

Downloading and Verifying an FTP File Using FileZilla Client

Objective

To download the previously uploaded image file from the FTP server to a different folder and verify that the downloaded file size matches the original file.

Step 1: Open FileZilla Client

1. Open **FileZilla Client**.
2. Connect to the FTP server using the credentials created earlier.

Example:

Host: 192.168.1.10

Port: 21

Username: ftpuser

Password: [your configured password]

Step 2: Locate the Uploaded File

After connecting:

1. Look at the **Remote site** panel on the right side.
2. Open the FTP shared folder.
3. Locate the uploaded image file, for example:

sample.jpg

Step 3: Select a Different Local Folder

In the **Local site** panel on the left side:

1. Navigate to a different folder from the original location.
2. For example, create or select:

C:\Users\Student\Downloads\FTP_Test

This ensures that the downloaded file is saved separately from the original.

Step 4: Download the File

1. Right-click sample.jpg in the **Remote site** panel.
2. Select **Download**.
3. Wait for the transfer to complete.
4. Check the **Successful transfers** section to confirm that the download was successful.

Step 5: Verify the File Size

Open the folder containing the original image and the folder containing the downloaded image.

In Windows:

1. Right-click the original file.
2. Select **Properties**.
3. Record the **Size**.

4. Right-click the downloaded file.
5. Select **Properties**.
6. Compare the two sizes.

Example:

File	Location	Size
Original sample.jpg	C:\Users\Student\Pictures	2.45 MB
Downloaded sample.jpg	C:\Users\Student\Downloads\FTP_Test	2.45 MB

If both sizes are identical, the file was transferred correctly.

Step 6: Optional Command-Line Verification

The file size can also be checked using PowerShell:

```
(Get-Item "C:\Users\Student\Pictures\sample.jpg").Length
```

```
(Get-Item "C:\Users\Student\Downloads\FTP_Test\sample.jpg").Length
```

The two numbers should be identical because they represent the file size in bytes.

Task-4

Here are three real-world scenarios where Simple Network Management Protocol (SNMP) is widely utilized to maintain network health, uptime, and performance:

1. Data Center Environmental & Power Monitoring

- **Context:** Large cloud providers and enterprise data centers housing thousands of servers in racks.
- **Mechanism:** Network Operations Centers (NOCs) use SNMP to poll Environmental Monitoring Units (EMUs), smart Power Distribution Units (PDUs), and Uninterruptible Power Supplies (UPSs).
- **Use Case:** Sensors track ambient rack temperature, humidity, and airflow. If a cooling unit fails and temperatures spike, the PDU triggers an **SNMP Trap** (an asynchronous alert) to the management platform, instantly notifying engineers to prevent hardware thermal shutdowns.

2. Enterprise Multi-Location VPN and Branch Router Monitoring

- **Context:** Retail chains or corporate networks connecting dozens of remote branch offices to a central headquarters.

- **Mechanism:** Central network management systems (like Nagios or Zabbix) use SNMP polling at regular intervals to query branch routers for CPU load, memory utilization, interface error counters, and IPsec tunnel states.
- **Use Case:** If a branch office's site-to-site VPN drops or packet loss spikes due to an unstable carrier line, the SNMP monitoring tool flags the degradation, enabling IT staff to proactively troubleshoot or failover to a backup broadband connection.

3. Telecom Cell Tower and Mobile Backhaul Management

- **Context:** Telecommunication companies managing thousands of remote cellular base stations (cell towers).
- **Mechanism:** Base transceiver stations and microwave backhaul links are equipped with SNMP agents that report operational metrics back to the centralized network operations center.
- **Use Case:** Operators monitor battery backup voltage levels during grid power outages, physical door intrusion alarms on equipment cabinets, and radio link bandwidth capacity to ensure uninterrupted cellular and mobile data services.

Task-5

FTP vs HTTP Comparison

Point	FTP (File Transfer Protocol)	HTTP (HyperText Transfer Protocol)
1. Port Number	Uses TCP port 21 for the control connection by default. (MDN Web Docs)	Uses TCP port 80 by default. (MDN Web Docs)
2. Security	Basic FTP is not encrypted , so usernames, passwords, and data can potentially be intercepted. Secure alternatives include SFTP or FTPS. (MDN Web Docs)	Basic HTTP is also unencrypted . HTTPS adds TLS encryption and is used for secure web communication. (MDN Web Docs)
3. Data Transfer Method	Specifically designed to upload and download files between a client and server.	Uses request-response messages to transfer web pages, images, videos, files, and API data. (MDN Web Docs)
4. Common Use Cases	Uploading website files, transferring files between computers, and managing files on an FTP server.	Browsing websites, accessing web applications, loading images/videos, and

Point	FTP (File Transfer Protocol)	HTTP (HyperText Transfer Protocol)
		communicating with web APIs. (MDN Web Docs)
5. Authentication	Commonly supports username and password authentication for accessing an FTP server.	Supports several authentication mechanisms, including Basic, Bearer, Digest, and others . (MDN Web Docs)
6. Typical Client	FileZilla Client, FTP command-line tools, and other FTP clients.	Web browsers such as Chrome, Edge, and Firefox, as well as API clients.
7. Communication Style	Uses separate control and data connections for file-transfer operations.	Uses client requests and server responses for individual resources. (MDN Web Docs)